



Two families of membrane potential waves in *B. subtilis* biofilms

Is c-di-AMP the missing driver?

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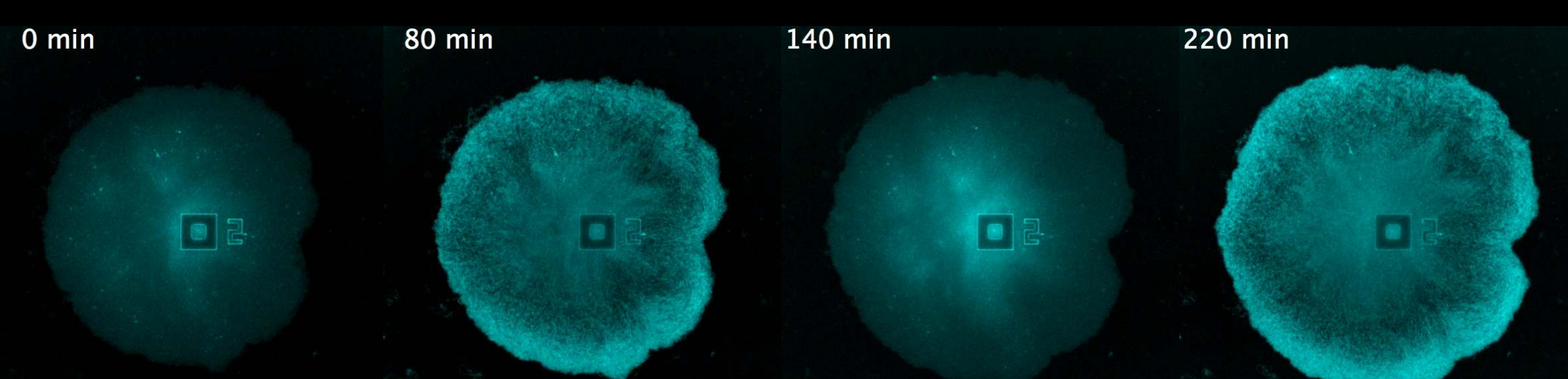
Abstract

In growing *B. subtilis* biofilms, we systematically observe a second, previously undescribed wave family: propagating fronts of cellular **depolarisation** (ThT-dark) that coexist with the classical **hyperpolarised** K^+ waves of Prindle et al. Existing models cannot explain them — they describe K^+ transport through a single channel (YugO), yet *B. subtilis* encodes five additional K^+ channels, all regulated by the small molecule c-di-AMP. **Is c-di-AMP the missing driver?** We address this question with three complementary approaches: (i) a minimal biophysical model coupling the K^+ wave to c-di-AMP-regulated channels and transporters; (ii) chemical and genetic perturbations of wild-type colonies on agar; (iii) a microfluidic chip for single-cell membrane-potential measurements. Together they aim to identify the molecular generator of depolarised waves and lay the groundwork for a quantitative description of electrical signalling in bacterial biofilms.

What is already known

In growing *B. subtilis* biofilms, long-range K^+ waves propagate across the colony and **hyperpolarise** every cell along their path. Imaged with the dye ThT (bright in hyperpolarised cells), each wave appears as a ThT-bright front coordinating metabolic stress and growth.

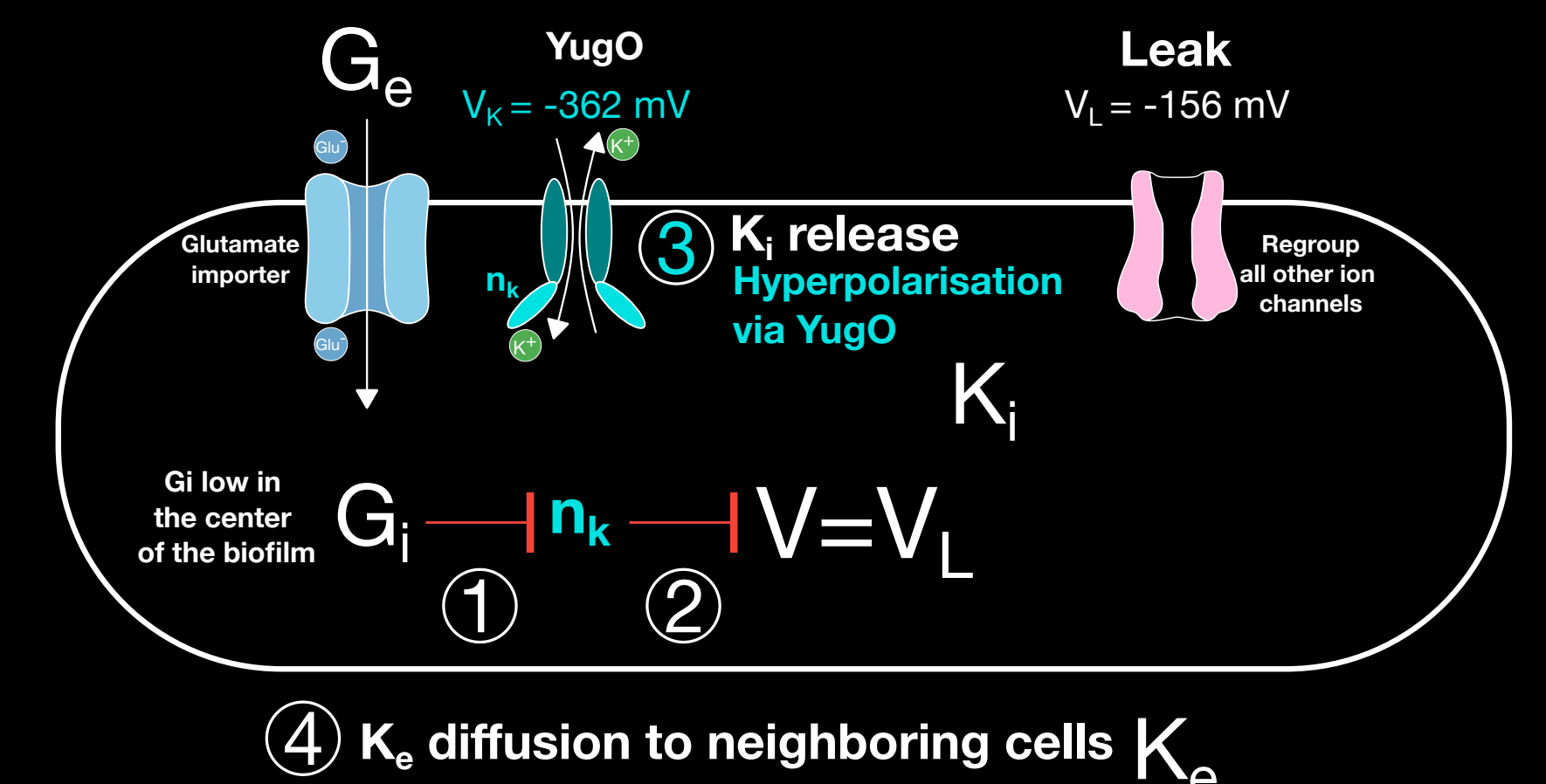
The Martinez-Corral model (2019) explains them through a single regulated channel, YugO, gated by G_i . When G_i drops, YugO opens (1); K^+ flows out (2), hyperpolarising the firing cell (3). The released K^+ diffuses (4) to neighbours and **transiently depolarises** them via V_L shift in the leak (5), which blocks their Glu uptake, drops their G_i (6), and opens their YugO (1) — propagating the **hyperpolarisation** wave.



Hyperpolarised waves propagating in *B. subtilis* biofilm

$V \downarrow$: Hyperpolarisation

$V \uparrow$: Depolarisation

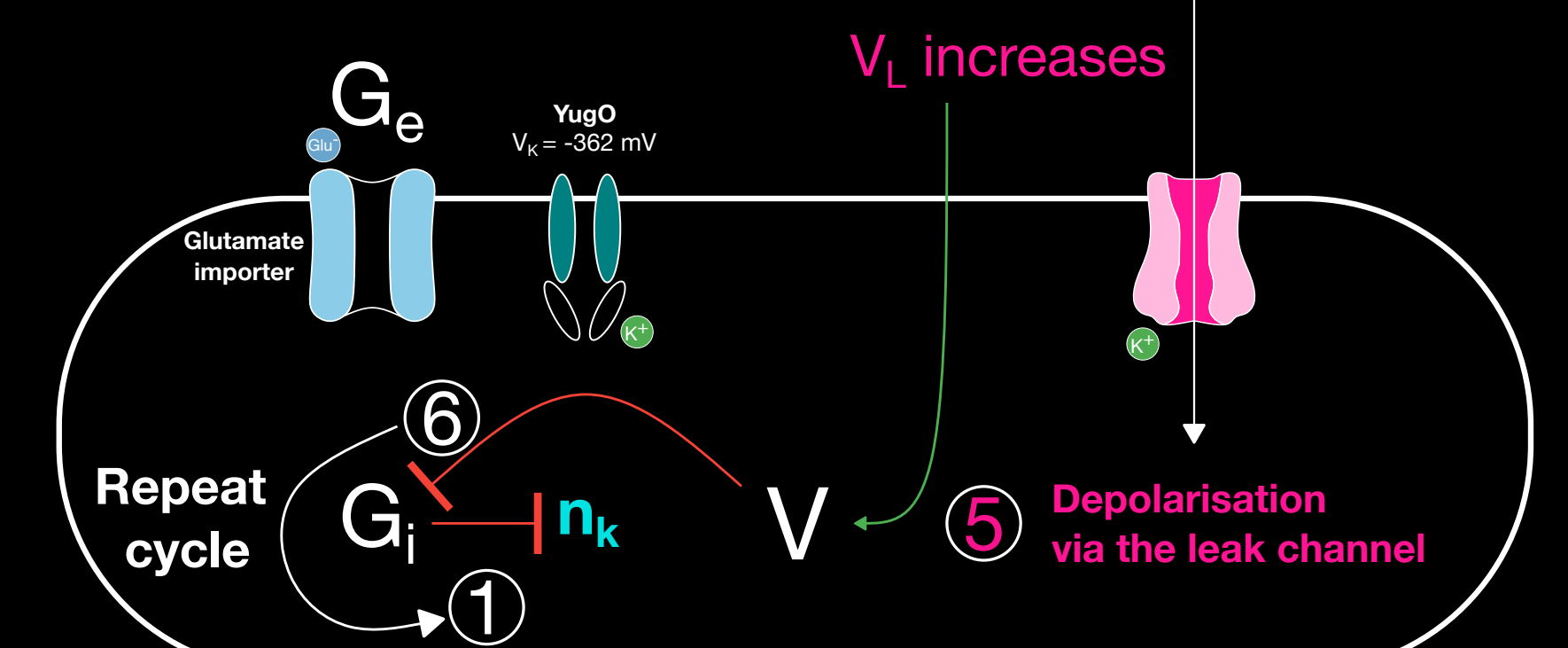


Legend

K: Potassium

G: Glutamate

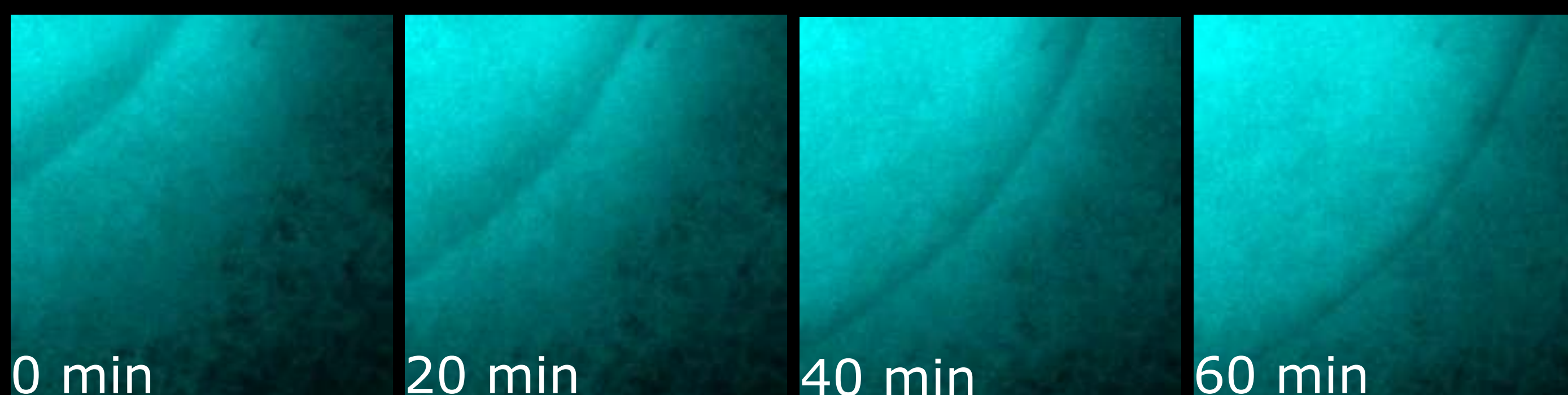
V: Membrane potential



Martinez-Corral model

A second wave family: depolarised waves

In every wild-type biofilm we image, propagating fronts of cellular **depolarisation** (ThT-dark) coexist with the **hyperpolarised** K^+ waves. This second wave family is not described in the literature and not produced by existing models.

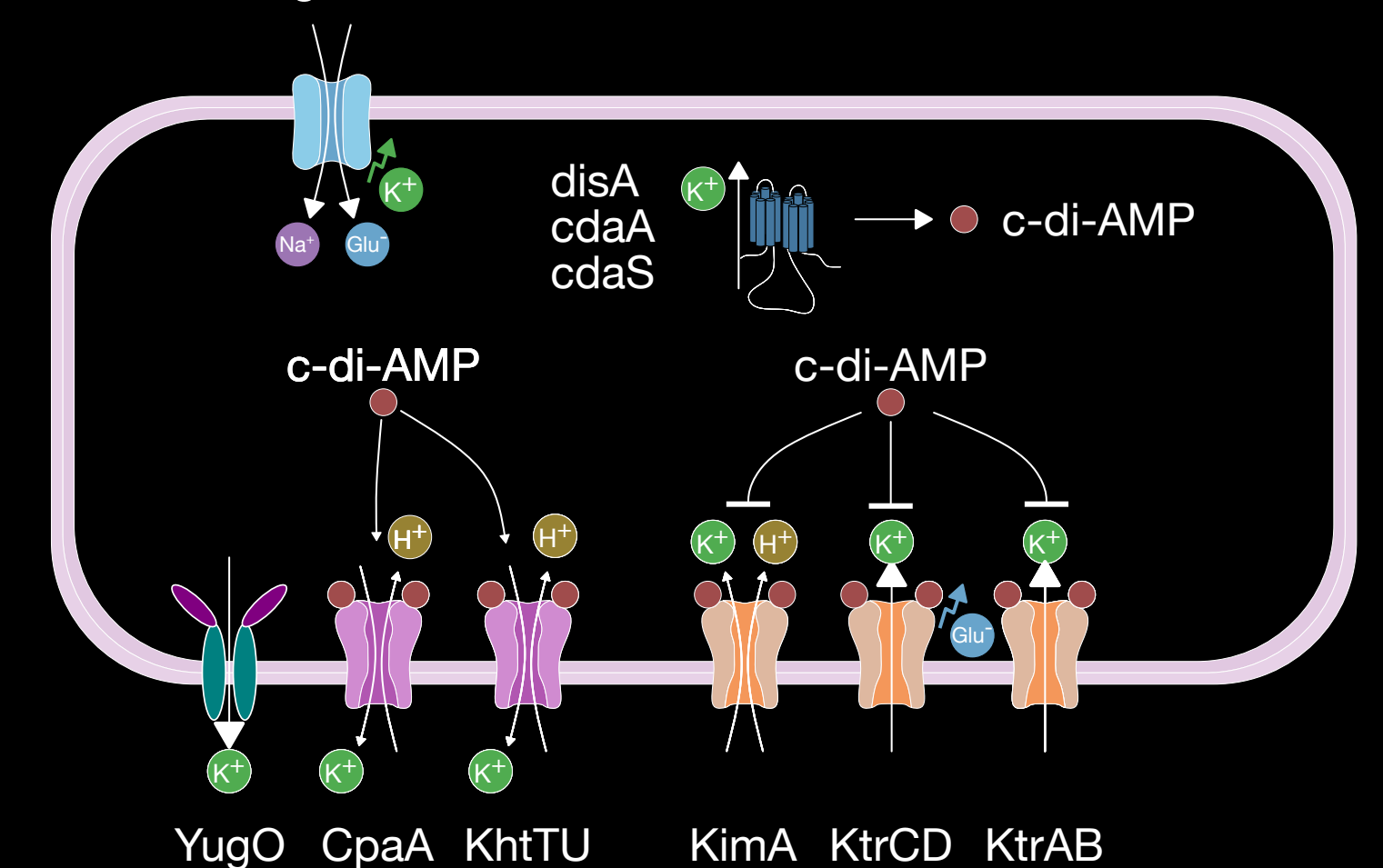


Depolarised waves propagating in *B. subtilis* biofilm

Does c-di-AMP drive depolarised waves?

B. subtilis encodes 6 K^+ transporters. YugO (the M-C channel) is the only one not regulated by c-di-AMP. The other 5 are inversely controlled: 3 importers closed, 2 antiporters opened.

Because intracellular K^+ raises c-di-AMP synthesis, K^+ accumulation triggers its own export.

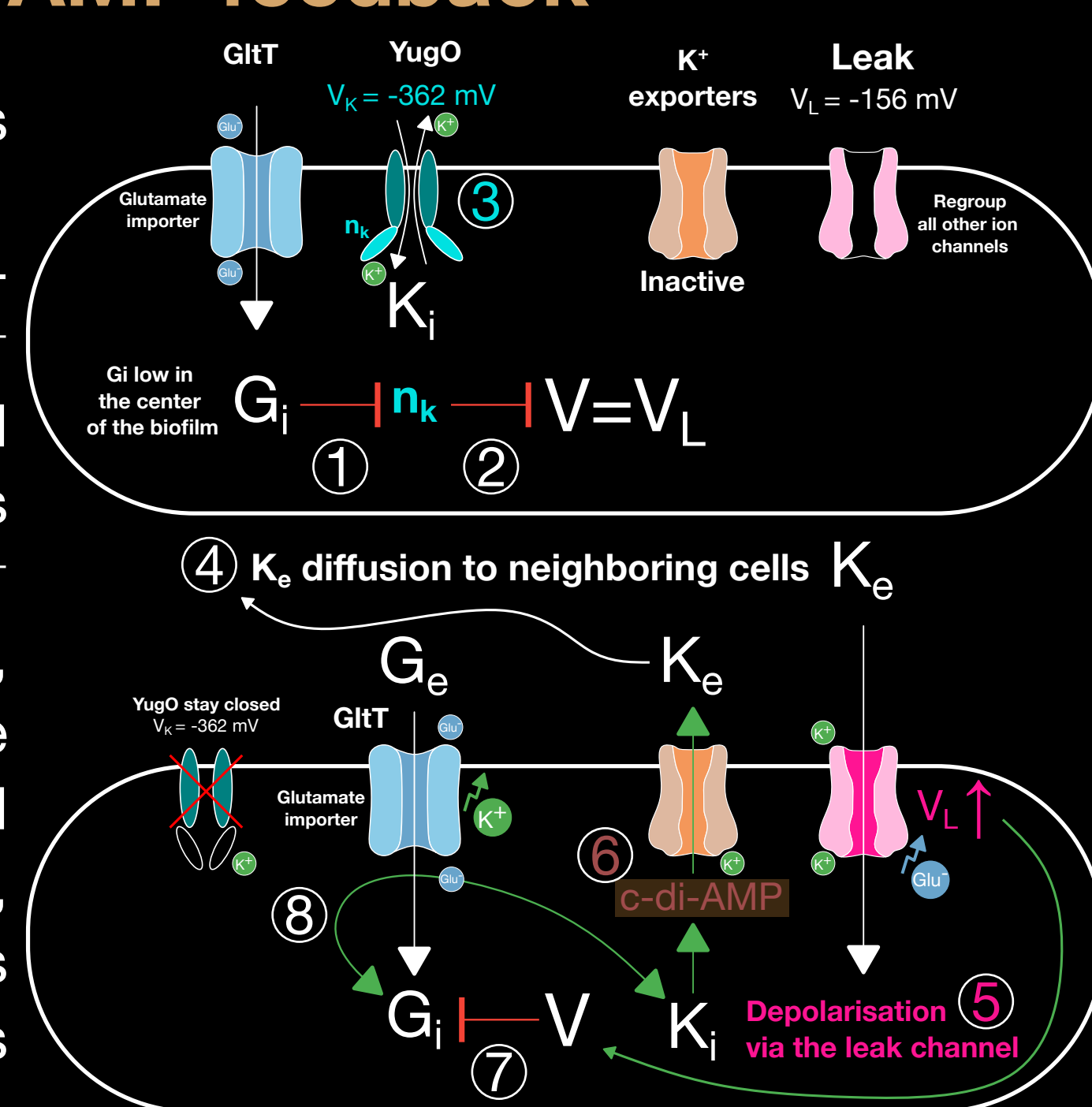


Could this feedback loop generate the depolarised waves?

Modeling: M-C model + c-di-AMP feedback

Top cell: same as M-C (steps 1–5, previous panel).

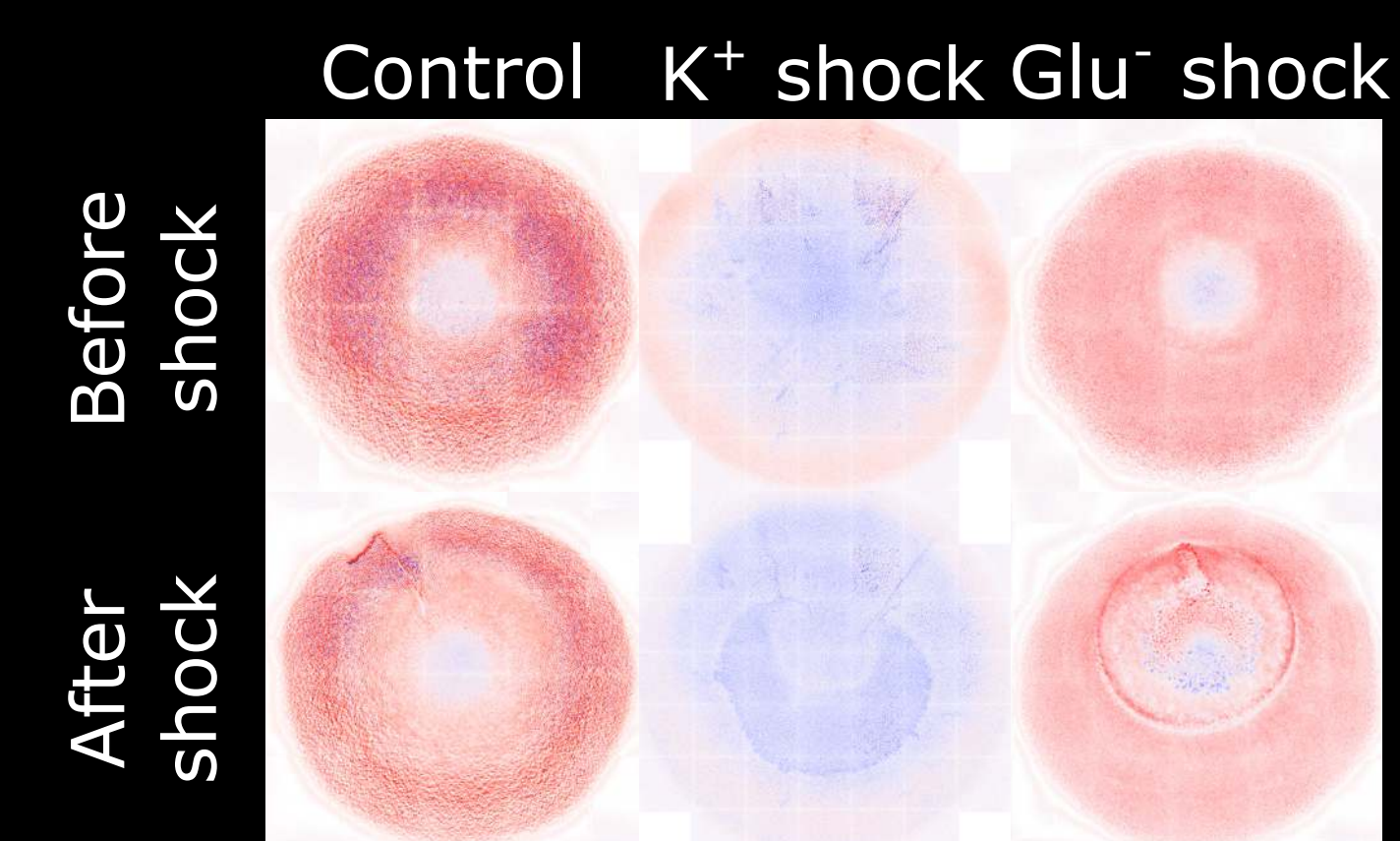
We propose two extra players: c-di-AMP-regulated K^+ exporters and GltT. K^+ entering via the leak **depolarises** V and raises K_i (as in M-C). But here, K_i activates c-di-AMP (6), which switches on K^+ exporters: K^+ is continuously pumped out, keeping K_e elevated and diffusing (4) to the next cell. Although V **depolarisation** would normally drop G_i (7), GltT keeps G_i high (8), so YugO stays closed. The cell stays sustainably **depolarised**, and its continuous K^+ release propagates the **dark wave**.



Agar: colony-scale perturbations

To confront our model with experiment, we apply chemical drops to **wild-type** biofilms on agar and image how each wave responds. Three drops tested: **water** (control), **glutamate**, **KCL**. Both glutamate and KCL shock produce visible, distinct effects on the **wave** dynamics — encouraging preliminary results.

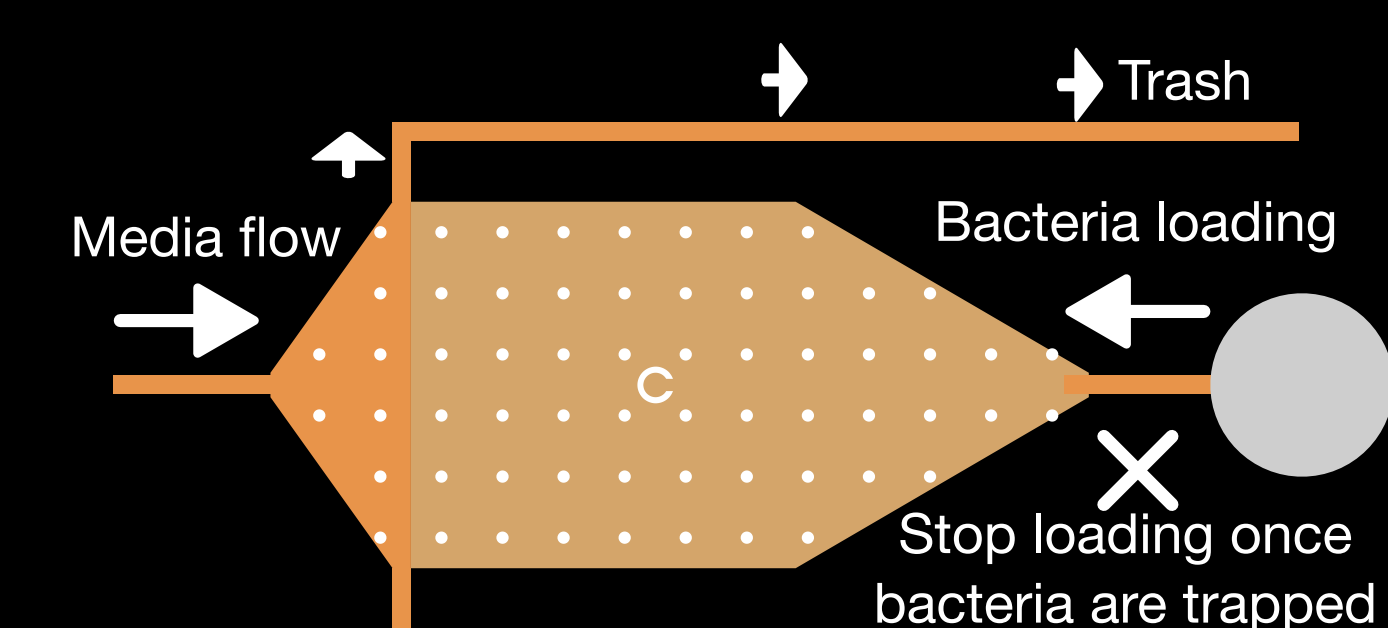
Next: do these effects persist in targeted K^+ transporter mutants?



Microfluidics: single-cell resolution

Why single-cell? Macroscopic waves are a collective effect: the same colony-scale pattern could in principle arise from several different single-cell behaviours. Only single-cell resolution can verify that the cell-by-cell mechanism we propose is what actually drives the observed waves.

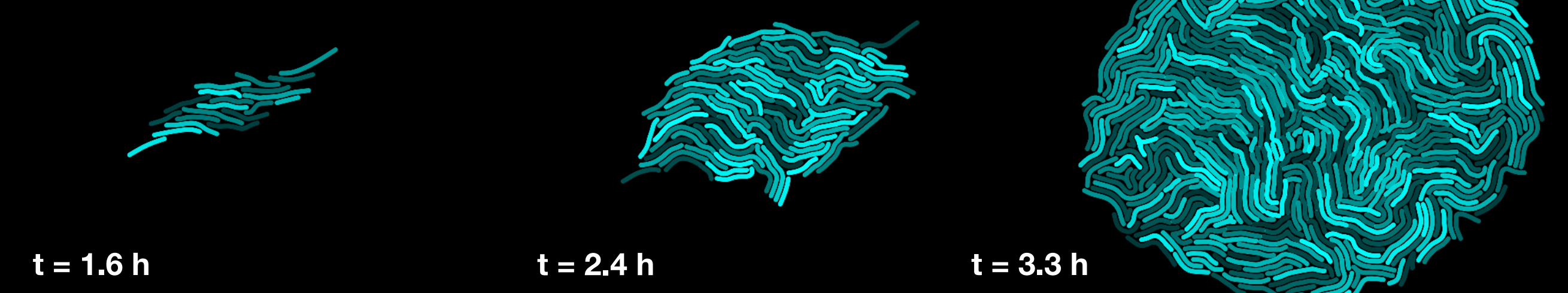
How? A custom two-layer PDMS chip traps *B. subtilis* in a 1.2 μm monolayer — a strictly 2D biofilm — while media perfuses through the 6 μm layer above. This monolayer is the key: it enables clean cell-by-cell segmentation at 100 $\times$ (impossible in a multilayer biofilm). Dual imaging at 4 $\times$ (colony) and 100 $\times$ (cells) gives the membrane-potential trajectory of every cell during a wave — directly comparable to the model.



1.2 μm layer
6 μm layer
C Cell trap (incoming reasearh)

Towards a digital biofilm

We will build a digital biofilm — thousands of rod-shaped cells dividing, growing, each carrying its own membrane-potential dynamics. Confronted cell-by-cell with the microfluidic data, this in-silico colony becomes the strongest test of the proposed mechanism, and the first agent-based model of a biofilm with cell-resolved electrical dynamics.



References:

Prindle et al. (2015) Nature · Liu et al. (2015) Nature · Martinez-Corral et al. (2018) PNAS, (2019) Phil Trans R Soc B · Larkin et al. (2018) Cell Syst. · Lopez et al. (2009) PNAS · Mehne et al. (2014) JBC · Townsley et al. (2018) mBio · Gundlach et al. (2019) JBC · Krüger et al. (2020) J Bact., (2021) JBC · Chou et al. (2022) Cell

